Independence

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1 Definitions

First, an example

Flip a fair coin twice. Let A be the event "the first flip is heads" and B be "the second flip is heads." Each has probability $P(A) = P(B) = 1/2$, and the joint event $A \cap B$ ("both heads") has probability $P(A \cap B) = 1/4 = P(A)\,P(B)$, so A and B are independent — as our intuition demands, since the two flips do not influence each other.

A subtler case: let C be "exactly one flip is heads." Then $P(C) = 1/2$, and $P(A \cap C) = P(\text{first is H, second is T}) = 1/4 = P(A)\,P(C)$, so A and C are *also* independent — even though C explicitly references the outcome of the first flip. Independence is a numerical condition on probabilities, not a statement about causal or logical entanglement.

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

Definition 1 (Pairwise independence)**.** *Events $A, B \in \mathcal{F}$ are* independent *iff*

$$P(A \cap B) = P(A)\,P(B).$$

Read aloud: *the probability of A intersect B equals the probability of A times the probability of B.*

Definition 2 (Mutual independence)**.** *A finite family $\{A_1, \ldots, A_n\} \subset \mathcal{F}$ is* mutually independent *iff for every non-empty subset $S \subseteq \{1, \ldots, n\}$,*

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Read aloud: *the probability of the intersection of the A-sub-i for i in S equals the product over i in S of the probability of A-sub-i. Mutual independence is strictly stronger than pairwise independence.*

Definition 3 (Conditional independence)**.** *Given $C \in \mathcal{F}$ with $P(C) > 0$, events A, B are* conditionally independent given C *iff*

$$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C).$$

Read aloud: *the probability of A intersect B given C equals the probability of A given C times the probability of B given C.*

2 Main theorem

Theorem 1 (Independence of complements)**.** *If A and B are independent, then so are A^c and B.*

Proof. Decompose B as a disjoint union $B = (A \cap B) \sqcup (A^c \cap B)$. By finite additivity,

$$P(B) = P(A \cap B) + P(A^c \cap B),$$

Read aloud: the probability of B equals the probability of A intersect B plus the probability of A-complement intersect B. so

$$P(A^c \cap B) = P(B) - P(A \cap B).$$

Read aloud: the probability of A-complement intersect B equals the probability of B minus the probability of A intersect B. Using independence of A and B,

$$P(A^c \cap B) = P(B) - P(A)\,P(B) = (1 - P(A))\,P(B) = P(A^c)\,P(B).$$

Read aloud: the probability of A-complement intersect B equals the probability of B minus the probability of A times the probability of B, which equals one minus the probability of A, all times the probability of B, which equals the probability of A-complement times the probability of B. Hence A^c and B are independent. $\square$

Applying the theorem twice (first to (A, B), then to (B, A^c)) shows that A^c and B^c are also independent.

3 A pairwise-but-not-mutually-independent triple

Example 1. *Let $\Omega = \{1, 2, 3, 4\}$ with the uniform measure. Define*

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

Read aloud: *A equals the set containing 1 and 2; B equals the set containing 1 and 3; C equals the set containing 1 and 4. Then $P(A) = P(B) = P(C) = \frac{1}{2}$. Each pairwise intersection equals $\{1\}$, so $P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, giving pairwise independence. But $A \cap B \cap C = \{1\}$, so $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$.*